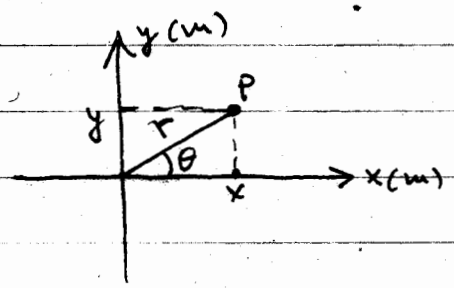
S&F, ch.1
Starych

Physics 1100
Solutions to HW02

P.35

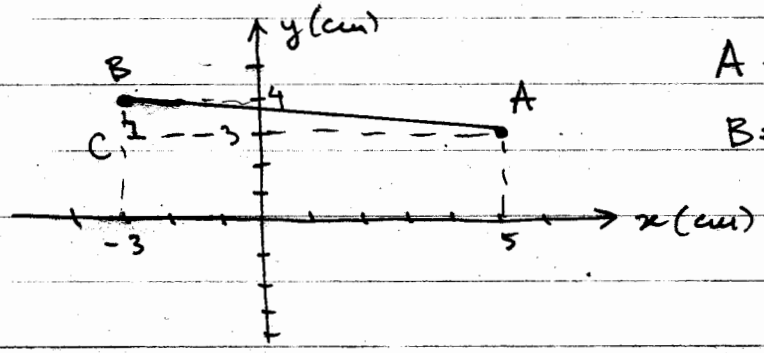


$r = 2.5\,m$, $\theta = 35°$

$x = r\cos\theta = 2.5 \cdot \cos 35° = 2.05 \approx \boxed{2.1\,m}$

$y = r\sin\theta = 2.5 \cdot \sin 35° = 1.43 \approx \boxed{1.4\,m}$

P 38



$A = (5.0, 3.0)$ cm

$B = (-3.0, 4.0)$ cm

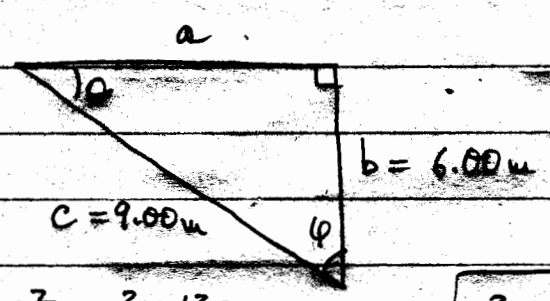
Find $|AB|$?

Construct triangle ABC by adding point $C = (-3.0, 3.0)$ cm.
Side CA is parallel to axis x, side CB is parallel to y
$\Rightarrow$ angle $\widehat{BCA} = 90°$.
length of side CB is 1.0 cm; length of CA is 8.0 cm.

$$\Rightarrow |AB|^2 = |CB|^2 + |CA|^2 = (1.0\,cm)^2 + (8.0\,cm)^2 = 65.0\,cm^2$$

$$\Rightarrow |AB| = \sqrt{65.0}\,cm = 8.06\,cm = \boxed{8.1\,cm} = 8.1 \cdot 10^{-2}\,m$$

P39



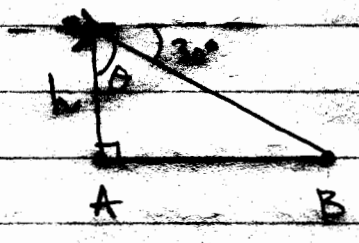
(a) $c^2 = a^2 + b^2 \Rightarrow a = \sqrt{c^2 - b^2} = \sqrt{91 - 36} = \sqrt{45} = 6.71\,m$

(b) $\left.\begin{array}{l} c\cos\theta = a \\ c\sin\theta = b \end{array}\right\} \Rightarrow \frac{b}{a} = \frac{c\sin\theta}{c\cos\theta} = \tan\theta \Rightarrow \tan\theta = \frac{6.00}{6.71} = 0.894$

(c) $c\cos\varphi = b$

$c\sin\varphi = a \Rightarrow \sin\varphi = \frac{a}{c} = \frac{6.71\,m}{9.00\,m} = 0.746$

P41



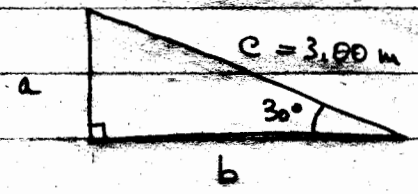
distance $|AB| = 6.00$ mi

altitude $h = ?$

Note that $\tan\theta = \frac{|AB|}{h} \Rightarrow h = \frac{|AB|}{\tan\theta}$

From the diagram, $\theta = 60° \Rightarrow h = \frac{6.00\text{ mi}}{\tan(60°)} = 3.46$ mi

P42



(a) side opposite the 30.0° angle is

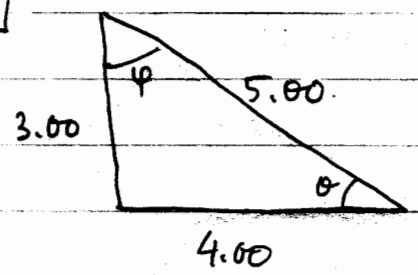
$a = c\sin 30° = 3.00 \cdot 0.5 = \boxed{1.50\text{ m}}$

(b) side adjacent to 30° angle

$b = c\cos 30° = 3.00 \cdot 0.866 = 2.598 =$

$= \boxed{2.60\text{ m}}$

P43



(a) 3.00 m side

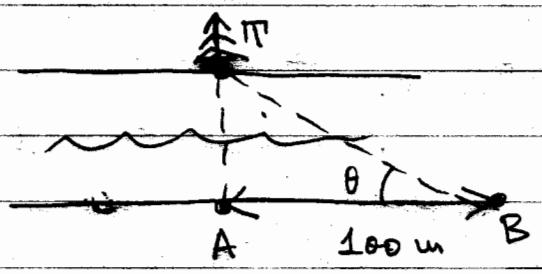
(b) 3.00 m side

(c) $\cos\theta = \frac{4.00}{5.00} = 0.800$

(d) $\sin\varphi = \frac{4.00}{5.00} = 0.800$

(e) $\tan\varphi = \frac{4.00}{3.00} = 1.333$

P46



$|AB| = 100$ m , $\theta = 35.0°$

Triangle ABT: $\tan\theta = \frac{|AT|}{|AB|}$

$\Rightarrow |AT| = |AB| \tan\theta = 100 \cdot 0.700 = 70$ m